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PRODUCT GUIDE: STRAIGHT LINE WIPERS

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GENERAL INFORMATION AND SECURITY SUMMARY

Safety Precautions

WARNING!

BE AWARE OF POTENTIAL INJURIES!

BE SURE TO READ AND FOLLOW THE INSTRUCTIONS BELOW BEFORE WORKING ON THE WIPER SYSTEM!

If the complete windshield wiper systems are to be installed by the customer, we do not have any right to supervise or intervene in the installation process. Therefore, we cannot be held responsible for any issues arising from installation errors. In cases where installation or maintenance instructions are insufficient or if any specific problems occur, please contact KARACA GEMİ Customer Service directly.

Most wiper motors feature a park setting that allows the motor to move to the park position through the vehicle's electrical system, even when the wipers are in the "off" position. For this reason, **the wiper arm should not be installed during this time, and no one should have their hands, fingers, or any body parts near the system while the wiper system is operating.**

Even small wiper motors cannot be stopped or braked by hand.

Never approach the area where the drive belt is located while the system is operating!

When connecting the wiper motor to the ship's electrical system, falling objects may cause injuries even if the wiper switch is in the off position. Therefore, **objects such as screwdrivers must not be left in the wiper system installation area.**

Ensure that the equipment is handled carefully. **Do not drop or strike** the equipment against hard surfaces. Do not apply any impact to the equipment during installation; otherwise, the wiper system may become mechanically deformed, which can lead to premature failure of the wiper mechanism.

1. INTRODUCTION

This user manual has been prepared to introduce the windshield wiper mechanism used on ships and to ensure its correct usage. The mechanism in question is a specialized system developed for cleaning ship windows and holds critical importance in the maritime sector for providing safe, efficient, and uninterrupted visibility.



Figure 1. Straight Line Model

The ship windshield wiper mechanism is designed to clean all glass surfaces on the vessel, primarily to enhance crew safety during adverse weather conditions and sea voyages. Rain, seawater, and other contaminants can create visual obstructions on ship windows, making navigation more difficult. These wiper systems help ensure uninterrupted visibility for the crew by quickly removing water accumulation and dirt from the glass surfaces during navigation.

The wiping area of the wipers is designed and determined based on the standing position on the ship, the user's height, and the required field of vision. This approach ensures that the area to be cleaned is clearly defined and provides the ship operator with maximum visibility, enabling safe navigation.

2. WIPER DESCRIPTION AND FEATURES

Type C Straight Line Wipers are Heavy-Duty Straight Line Wipers equipped with an internally mounted electric motor. The wipers can be mounted either at the top or the bottom of the window. The mechanical specifications are listed in Table 1.

The main components of the windshield wiper system are:

1. Wiper casing
2. Wiper motor
3. Wiper arm
4. Wiper blade

Mechanical Features	
Wiper Case Sizes	165 x 255 x XXXX mm
Weight of 1 Meter Case	15 kg
Material	Aluminium
Motor Working Temperature Range	-30°C to +70°C

XXXX : Depends on the windows sizes

Table 1. Wiper Case Mechanical Properties

2.1. WIPER CASING

The length of the wiper casing is determined by adding a fixed value of 280 mm to the stroke distance (Figure 2). The minimum stroke length is set at 320 mm, resulting in a casing length of 600 mm. In a single-arm wiper system, when the maximum stroke length reaches 1720 mm, the casing length extends up to 2000 mm. By increasing the number of arms and motor mechanisms, the same operations can be applied to all windows, regardless of size. This configuration ensures flexibility and adaptability of the system by meeting the requirements for different stroke lengths.

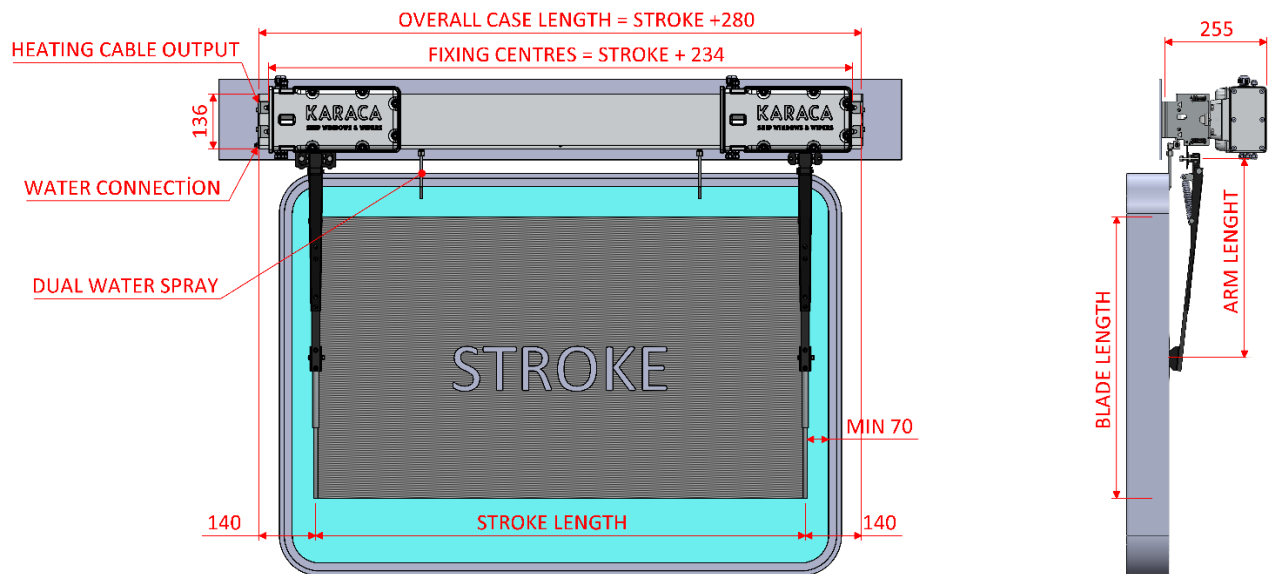


Figure 2. Wiper Casing General Overview

Wiper systems are designed and manufactured to be compatible with different types of glass (Figure 3). Wipers that are capable of adapting to flat, curved, or specially shaped windows provide effective cleaning performance on every surface. This variety meets the needs that vary according to the type of ship and operating conditions, enhancing both functionality and ensuring long-lasting use. The correct selection of wipers is critical to provide a safe field of vision while also preventing damage to the glass surface.

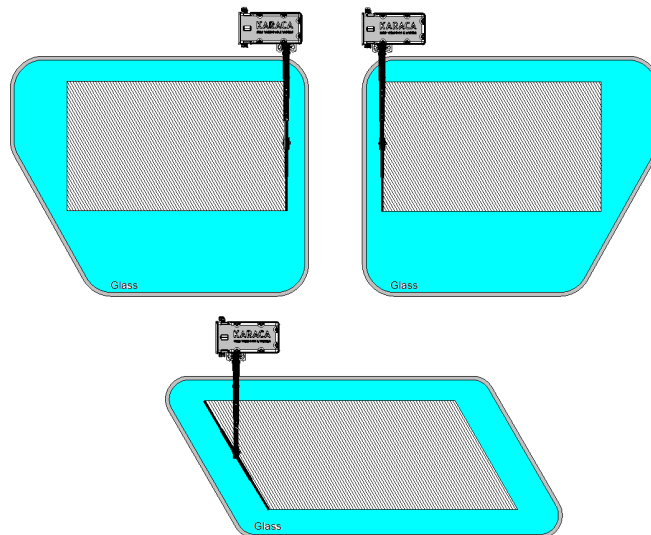


Figure 3. Wiper Types According to Different Glass Types

2.2. WIPER MOTORS

Straight Line Wipers operate with a 24VDC motor. The motor can be placed at both ends of the wiper casing by reversing the front cover. The park position can be adjusted to the right or left depending on the glass structure. All electric motors contain an infinite reduction gearbox. The windings have Class F insulation (Figure 4).

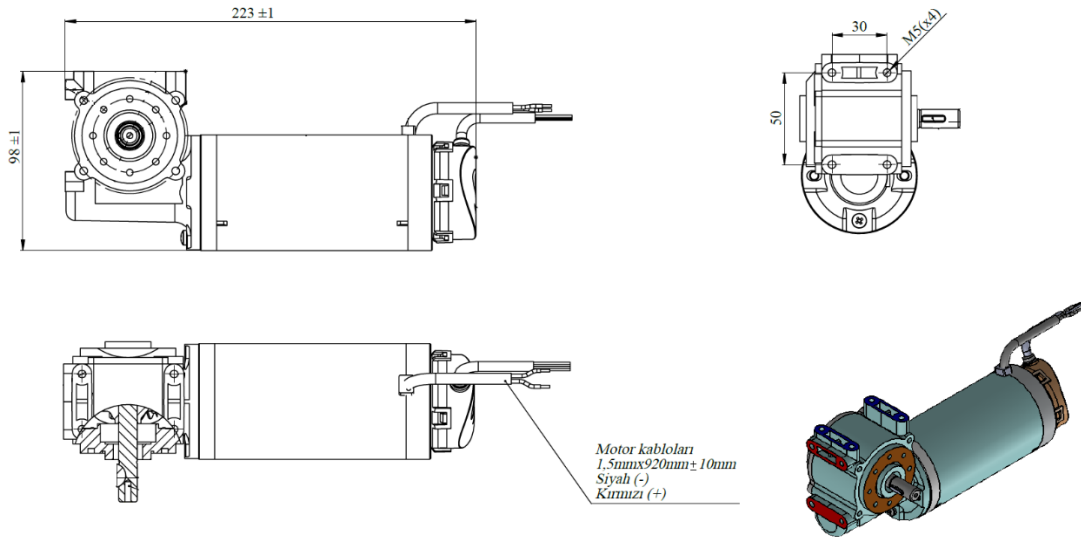


Figure 4. Straight Line Motors

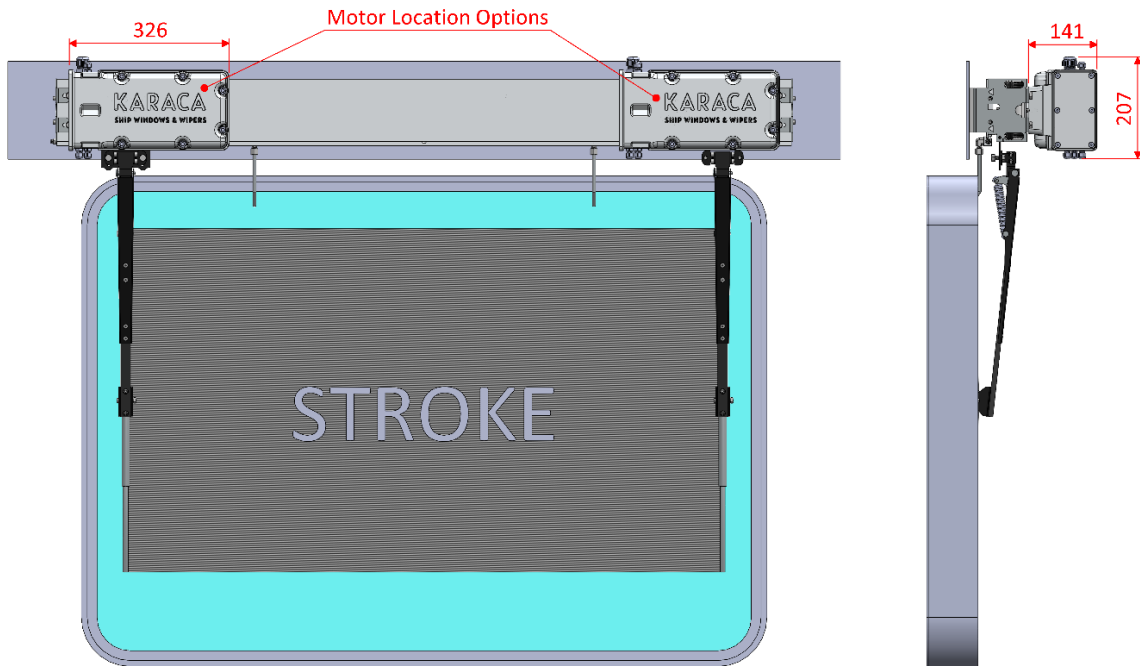


Figure 5. Motor Placing Options

The motor cover and technical dimensions are shown in Figure 5, and the external volume occupied by the motor outside the wiper casing is 141x207x326 mm.

The motors contain control boards, which are divided into two types. The older control unit operates at a single speed of 280 rpm. In motors with the new control unit, there is a two-speed option, offering 360 rpm and 280 rpm (Table 2).

	Old	New	
Step	1	1	2
Rpm	280	360	280
1 m/sec	1,1	0,85	1,1

Tablo 2. Datas of motor control cards

The 24 VDC motor draws 2 amps of current under normal operating conditions, but this value can rise to 8 amps under strain. In such cases, it is recommended to use a 10-amp fuse to protect the motor (Table 2). It is advised to install a separate fuse for each wiper. The fuses are not supplied by KARACA Marine but are installed by the ship. The new control boards include a fuse protection circuit, ensuring the system is protected. Motor specifications are provided in Table 3. The motor is resistant to water and dust with an IP 67 rating.

Nominal Voltage	Full load current at 50/60Hz	Fusing Value 50/60 Hz	Speed (m/sec)
24 VDC	2 Amper	8 Amper	0,85

Tablo 3. Motor Features

2.3. WIPER ARMS

The wiper arm lengths range from a minimum of 300 mm to a maximum of 1500 mm. This wide range offers flexible usage for different types of ships and wiper systems. The correct selection of arm length plays a critical role in creating an effective cleaning area for the wipers.

Wiper casings up to 2 meters can be operated with a single wiper arm. The number of arms is increased as needed or depending on the wiper position.

For wipers longer than 2 meters and up to 3 meters, a dual-arm system must be used (Figure 6).

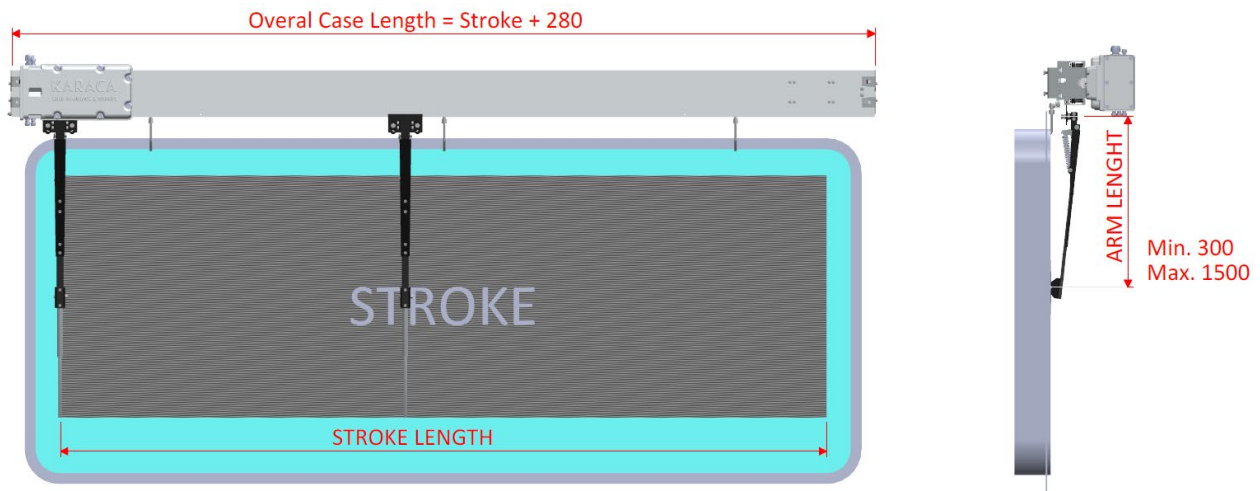


Figure 6. Dual-Arm Wiper System and Max-Min Arm Length

For wipers exceeding 3 meters, a three-arm system is preferred (Figure 7). This standard aims to enhance the functionality and reliability of the system, ensuring that windows of various lengths are cleaned with optimum performance.

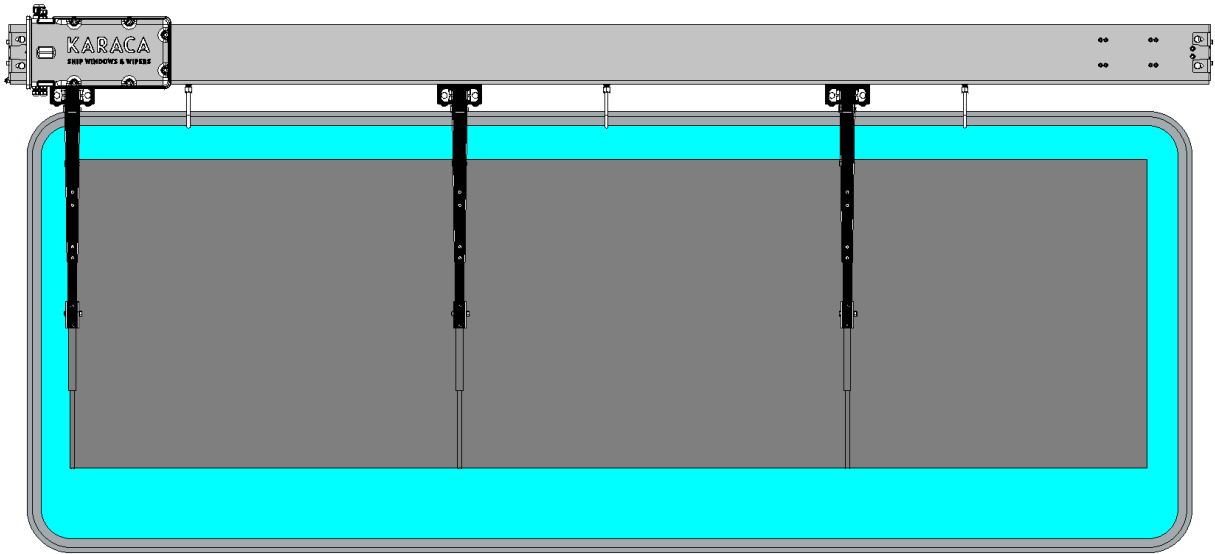


Figure 7. Example of a Three-Arm Wiper System

In cases where the angle between the wiper arm and the mounting part (the X angle in Figure 8) exceeds 20 degrees, the wiper arm must be bent. This is resolved by bending a lamella with dimensions of 10x20. These design details are crucial for both functionality and safety.



Figure 8. Wiper Arm Bending

2.4. WIPER BLADES

The wiper blade length ranges from a minimum of 300 mm to a maximum of 1200 mm. These dimensions are optimized to provide compatible wiper performance for different window frames. The correct selection of blade length enhances the efficiency of the cleaning area.

The angle of the blades is designed for a dual-blade structure with upper and lower wipers. It offers a total working angle of 46 degrees, with 15 degrees from the bracket and 31 degrees from the wiper slot (Figure 9). This design maximizes the movement range of the wipers, providing a wide field of vision.

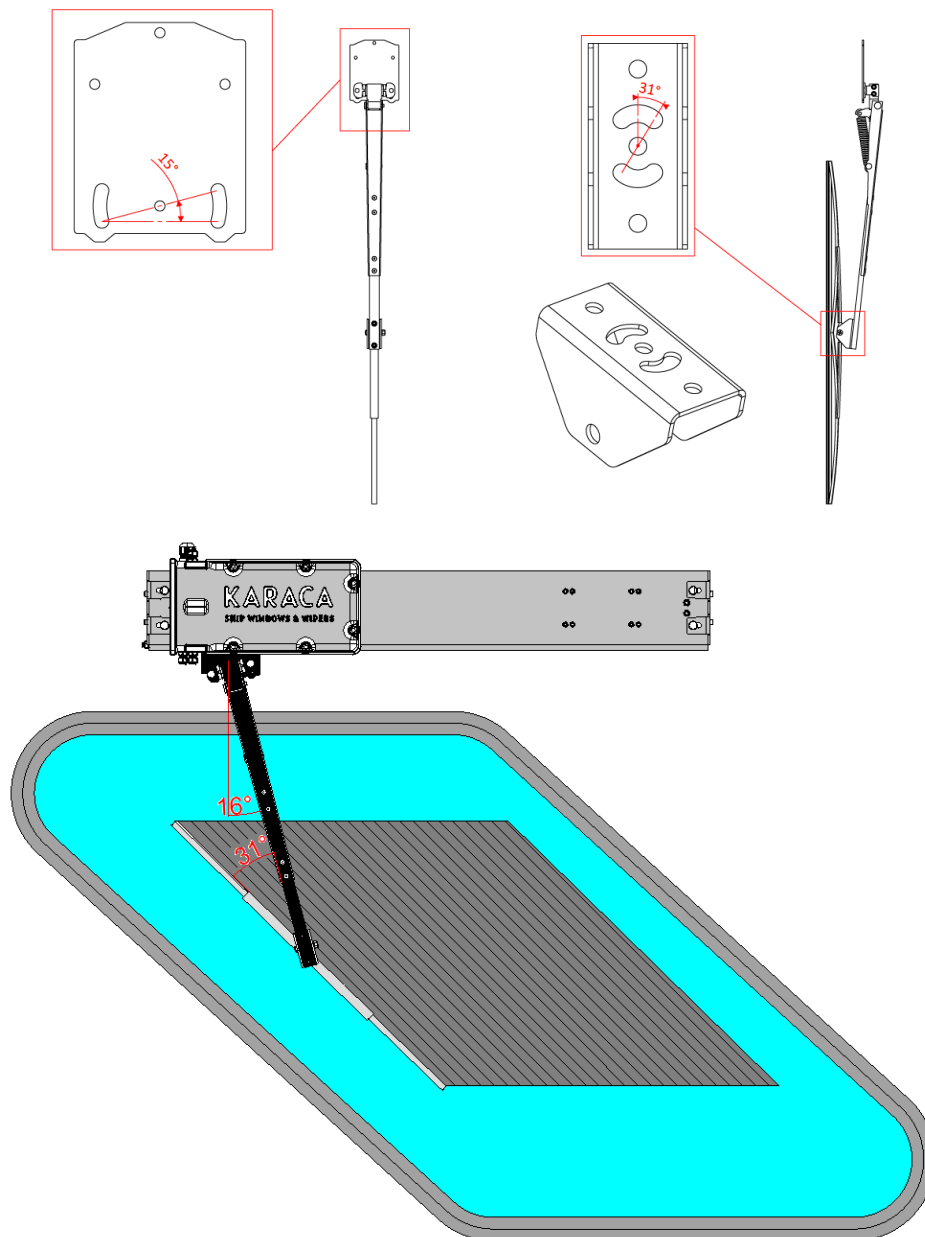


Figure 9. Wiper Blade Max Angle and 3D Image

The springs integrated into the wiper blades play a crucial role in applying optimal pressure toward the glass surface, enhancing cleaning efficiency. This design ensures that the blades make firmer contact with the glass surface, contributing to the effective removal of water, dirt, and other residues.

3. AUXILIARY PARTS

Auxiliary components are used to enhance the efficiency of the wiper mechanisms. Spray nozzles and heaters fall into this category.

3.1. SPRAY NOZZLES

The clean water source provides the spraying function through a compression connection with an 8mm general diameter for the hose (Figure 10) and a 6mm diameter for the pipe (Figure 11). The spray nozzles are specially adjusted for each window. For the wiper blades to perform effectively, they must be properly positioned, and their length must be correctly adjusted.

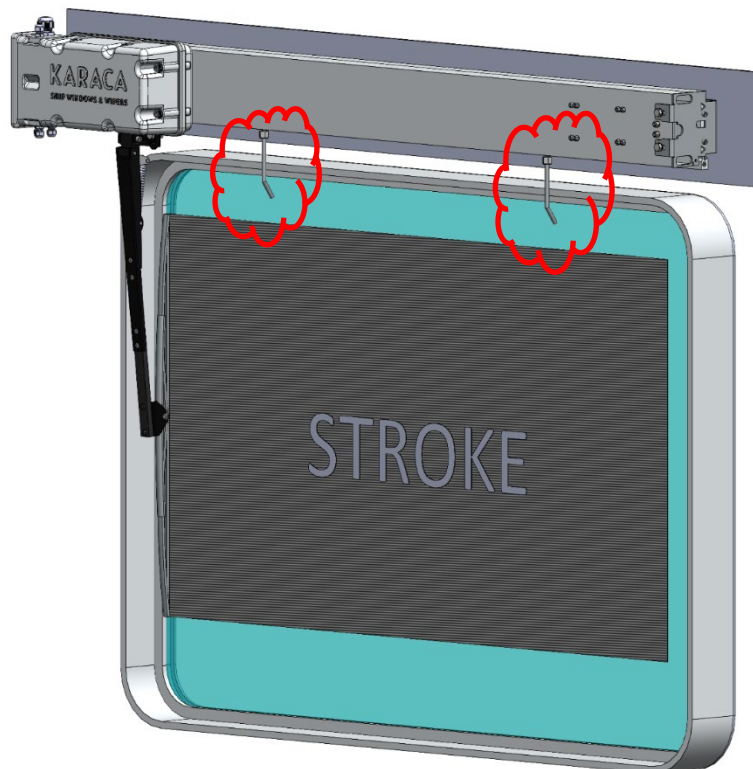


Figure 10. Pipe-Type Spray Component

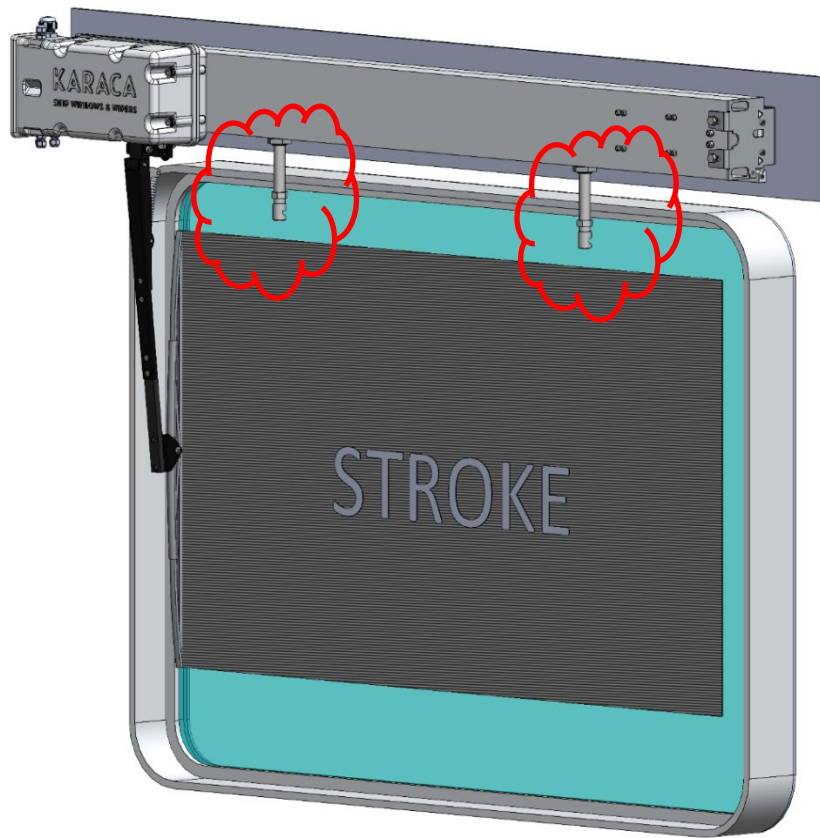


Figure 11. Nozzle-Type Spray Component

For windows with a stroke length under 1500 mm, two pipes or two nozzles must be used for each wiper. However, when the stroke length exceeds 1500 mm, the system configuration may vary depending on the window structure. In this case, three pipes or three nozzles must be installed for each wiper. Additionally, the shape and size of the glass frame can affect this arrangement, so the appropriate nozzle and pipe placement should be determined according to the characteristics of the glass frame for each application. These adjustments are crucial for the wiper system to function effectively and efficiently.

The installer must provide a pressurized water source and interconnection piping. When the washing option is installed, the maximum pressure for the system is 8 bar or 118 PSI, and the minimum pressure for adequate spraying reach is 1 bar or 15 PSI (Table 4). Example flow rates for a single spray jet are shown below.

Pressure		Flow Rate	
Bar	Psi	Litres/min	Gallons/min
1.0	15	0.95	0.20
1.5	22	1.20	0.25
2.0	29	1.40	0.30
3.0	44	1.75	0.40

Tablo 4. Water System Pressure and Flow Rate

3.2. HEATERS

To ensure the effective operation of the wiper system in cold weather conditions, optional de-icing heaters can be installed in the wiper casing. These heaters are designed to prevent the mechanism from freezing. The standard cable length is 3 meters, but longer cable options are also available upon request.

Power consumption varies depending on the stroke length of the wiper system. A 250W heater is used for a stroke length of 1 meter. The heating resistors operate at 220V.

For casings up to 2 meters, a single resistor is used, and the length of the resistor may vary. For casings longer than 2 meters, two resistors can be installed, one on the left side and one on the right.

Once the heaters are activated, they will remain on until manually turned off.

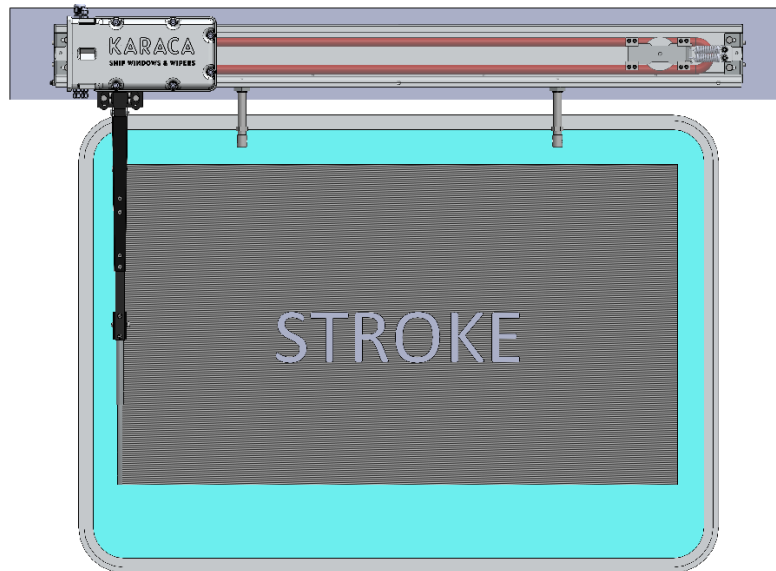


Figure 12. Resistor Appearance

4. WIPER INSTALLATIONS

Warning: Verify through the instruction form that the appropriate wiper, blade, and arms are selected for each window.

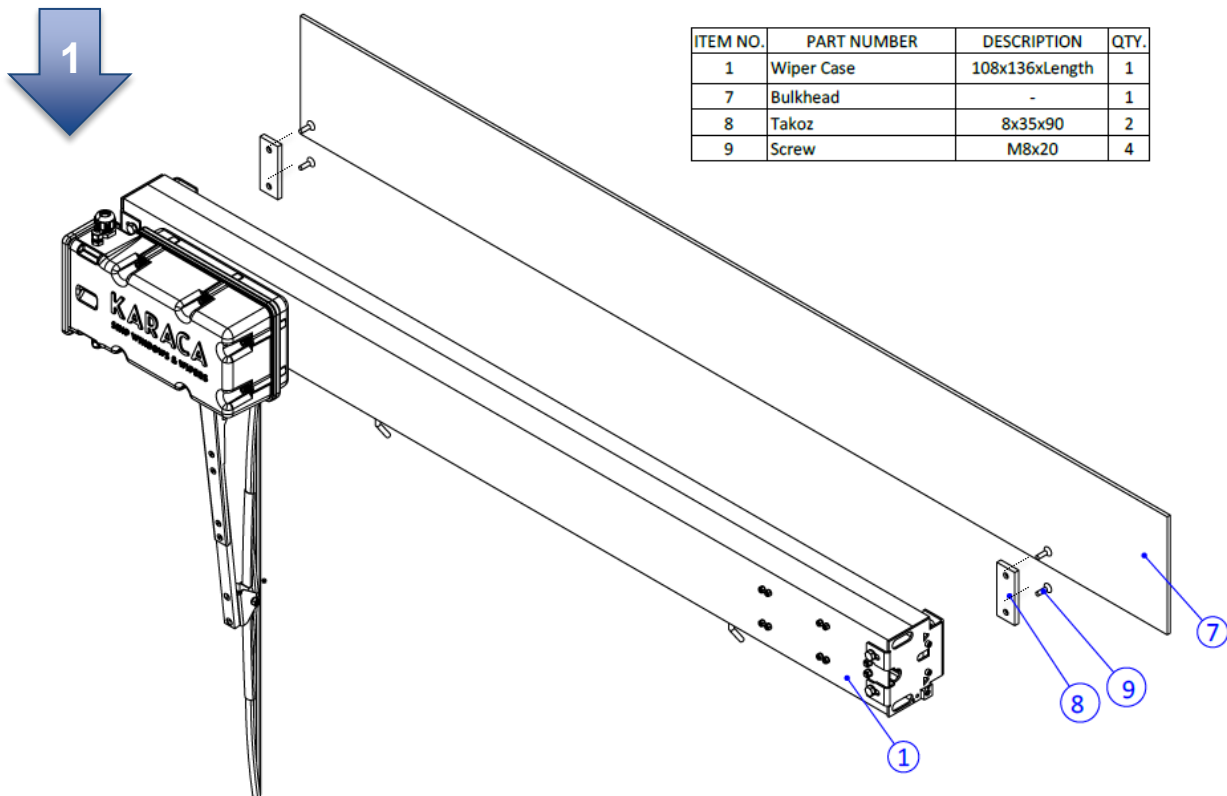
Warning: Before starting the installation, check that there are no obstructions or hazards in the selected location. The main frame should not obstruct the wiper's proper operation, and the body should be mounted on a flat surface to prevent bending.

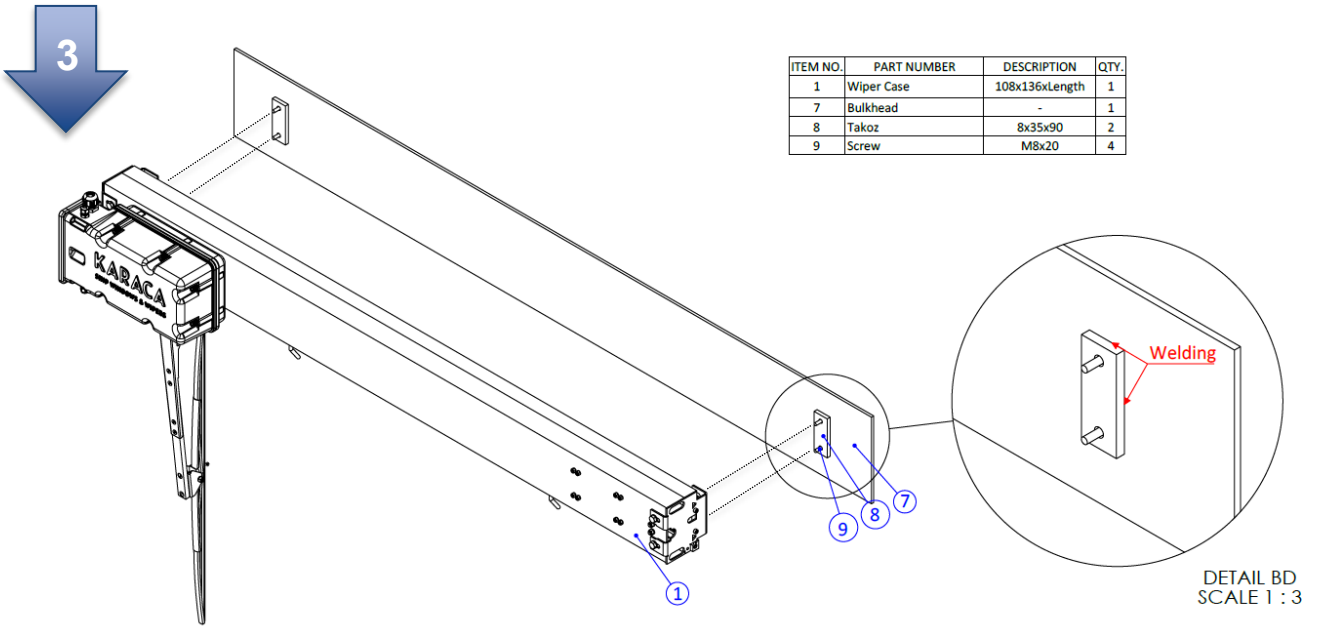
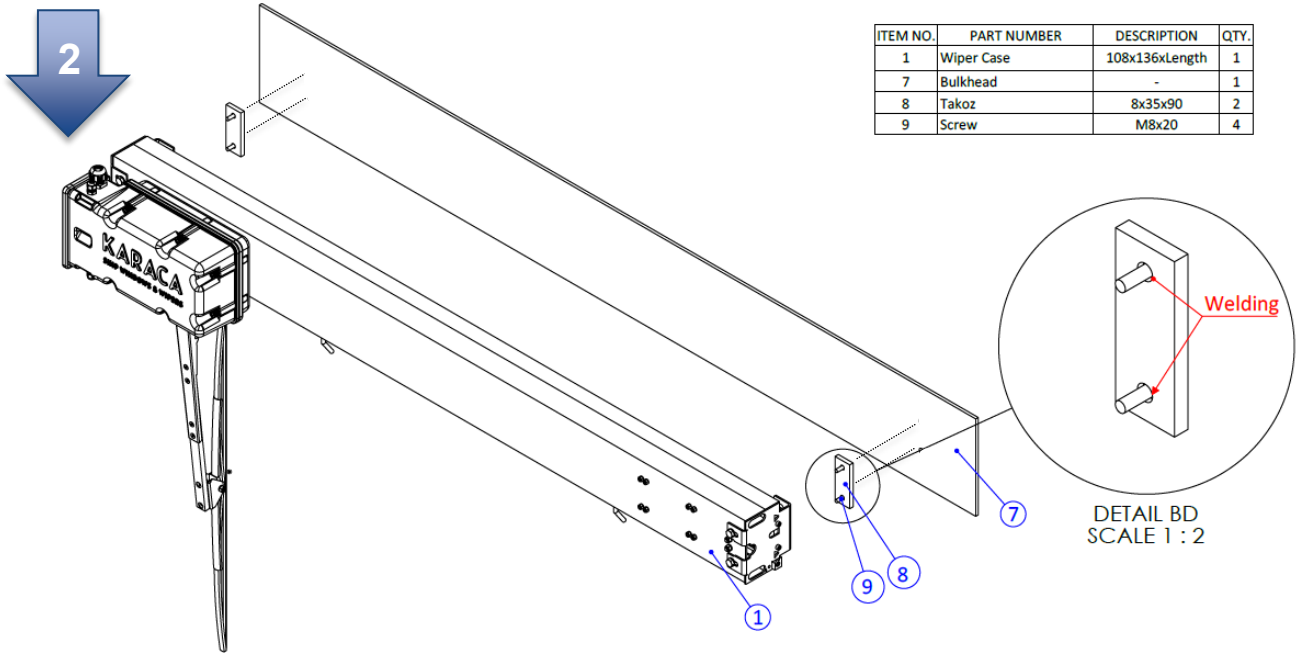
Warning: If multiple wiper units are to be installed in close proximity, ensure that there is at least an 80 mm distance between the units.

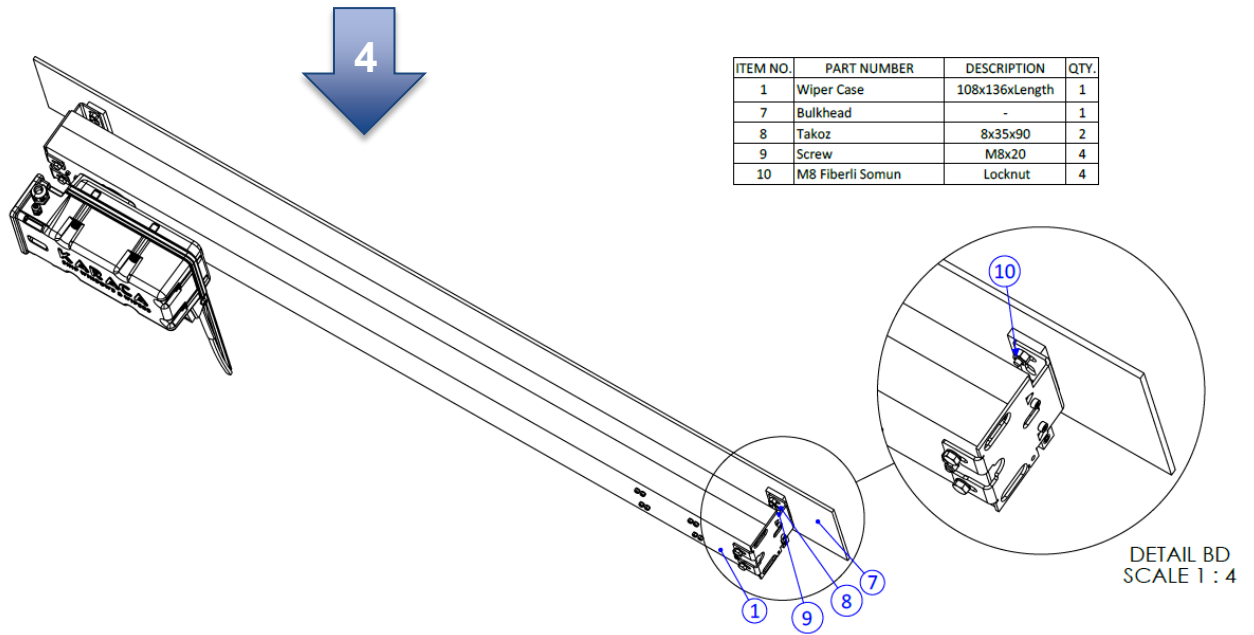
On the ship's hull:

Drill cable holes for the heater and motor cables, ensuring all holes are round and free from burrs. Apply a treatment to the bare metal to prevent corrosion. Ensure that the cables are sealed where they pass through the compartment. Use appropriate cable glands or seals to prevent water ingress and cable friction wherever the cables pass through the compartment.

Leave sufficient spare cable during the installation to allow the front assembly to be removed from the rear casing during maintenance.







1. The mounting kit includes a wedge (8) and M8 fiber nut (10).
2. The M8x20 bolt (9) is inserted into the wedge (8) and welded.
3. The wedge (8) piece is welded to the ship's hull (7).
4. The wiper casing (1) mounting holes (Figure 13) are placed over the bolts (9) on the wedge (8) and tightened with the fiber nut (10).

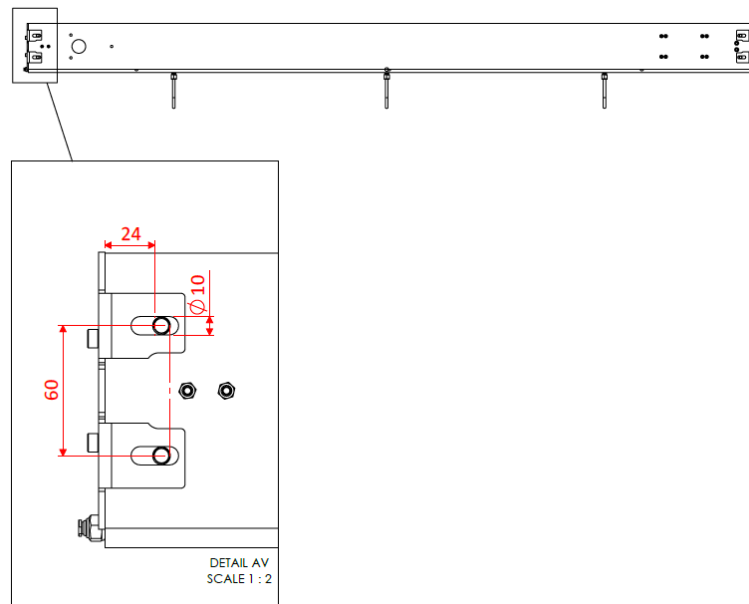


Figure 13. Wiper Casing Mounting Holes

Move the wiper arm assembly through its full stroke and check that the movement is not restricted (the motor may show some resistance, but the wiper should not be compressed). Investigate and correct any restrictions.

5. PART LIST

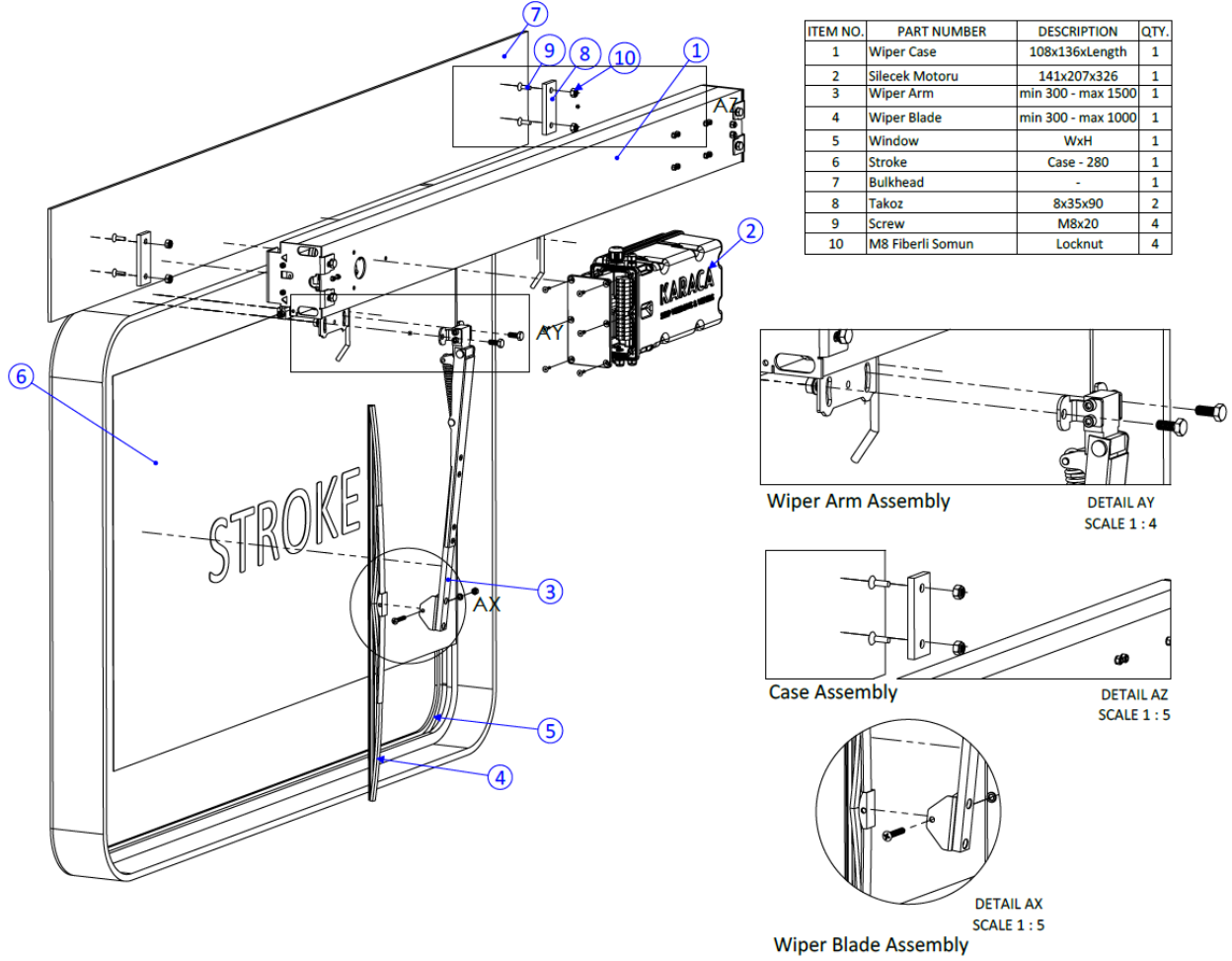
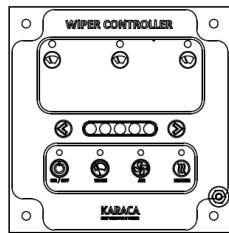


Figure 14. Part List

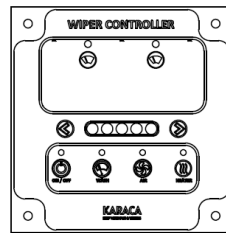
6. OLD-NEW CONTROL UNIT

6.1. GENERAL INFORMATIONS

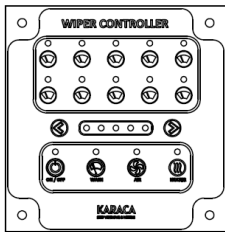
CT2N is a microcontroller-based wiper control system for up to 5 wiper motors; it provides all the functions for a proper windscreen cleaning. CT2N is composed by one (or more) control panels CP4PX (where 'X' stands for the number of wipers handled) and one relay box SRM2/3P or SRM4/5P depending on the number of wipers. Control panel is connected to relay box by one serial cable (CAN BUS CABLE) with two 8 pole plugs. Every unit has two plugs (in parallel) and any one or both can be used.



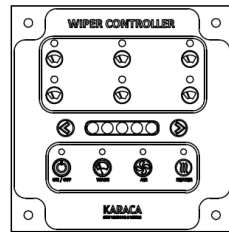
KRC-CP2-3way



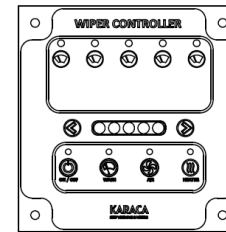
KRC-CP2-2way



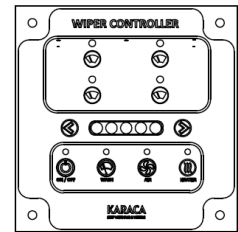
KRC-CP2-10way



KRC-CP2-6way



KRC-CP2-5way



KRC-CP2-4way

Features:

- ✓Double voltage supply: 12V - 24V
- ✓Activates/deactivates each wiper one by one
- ✓Synchronization in each selected mode
- ✓Three intermittent settings
- ✓Continuouns slow and fast mode
- ✓Perfect self parking position due to "dynamic brake"
- ✓Wipe/wash program
- ✓Activating the heater

✓Easy to install and to use

✓Reliable

Standard supply includes:

✓One KRCWCPX

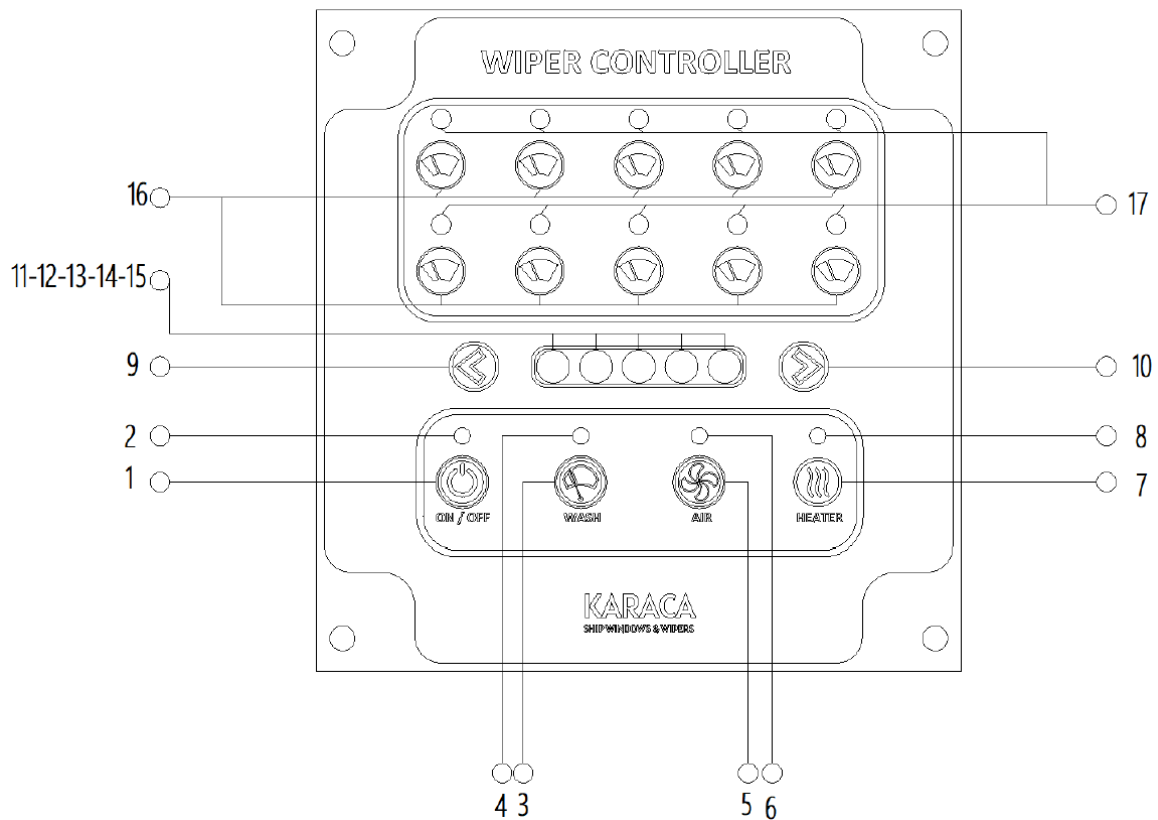
✓One KRCWB5

✓Two solenoid valve

✓One cable for communication (standard 5 meters, armored cat 6 cable)

✓Female connectors for motors connection to KRCWB5(one for each motor)

6.2. PUSH BUTTON FUNCTIONS AND SIGNALLINGS



With reference to fig 1:

1. General ON / OFF button
2. General ON / OFF and fault led

3. Washing system ON / OFF button
4. Washing system ON / OFF led
5. Air system ON / OFF button
6. Air system ON / OFF led
7. Heater ON / OFF button
8. Heater ON / OFF led
- 9-10. Decrease speed / increase delay in intermittent mode
- 11-12-13-14-15. Selected mode led bar
16. Single wiper ON / OFF button
17. Single wiper ON / OFF and fault led

6.3. OPERATING MODE

Available operating mode.

SPEED BAR	WAITING TIME	MODE
ALL LEDS OFF	-	STAND-BY
LED 11 ON	6 SECONDS	INTERMITTENT
LED 11+12 ON	4 SECONDS	INTERMITTENT
LED 11+12+13 ON	2 SECONDS	INTERMITTENT
LED 11+12+13+14 ON	-	CONTINUOUS LOW SPEED
LED 11+12+13+14+15 ON	-	CONTINUOUS HIGH SPEED

6.4. WIPE / WASH / AIR PROGRAM

By pressing the wash button, washing program will start.

| 3 sec. | basılı tut | 4 sec. | 2 sec. |

|=== Spray ===== Spray =====|

|=== Wiper low speed or high speed===|

|== Air ==|

*Press and hold the wash button to increase the washing time and spray/wipe time. When you release the wash button, automatic air will be activated. After the wipe/wash/air period the wipers will return to the selected park position.

6.5. FAILURES AND SIGNALLING

Wiper fault A flashing of a wiper led indicates that KRCWB5 detects an error on a wiper.

This may happen if:

- ✓ The parking switch is unconnected or damaged
- ✓ The motor doesn't run (internal self-resetting fuse in protection, motor unconnected or damaged)
- ✓ The motor doesn't run in FAST or in SLOW speed
- ✓ The motor can't be synchronized
- ✓ The motor connectors have been inserted after supplying power.

6.6. POWER SUPPLY

KRCWB5 will operate correctly if powered from 10V to 30V DC. Insert motor connectors BEFORE applying power supply. When powered, KRCWB5 will execute a lamp test and then will go in stand-by mode.

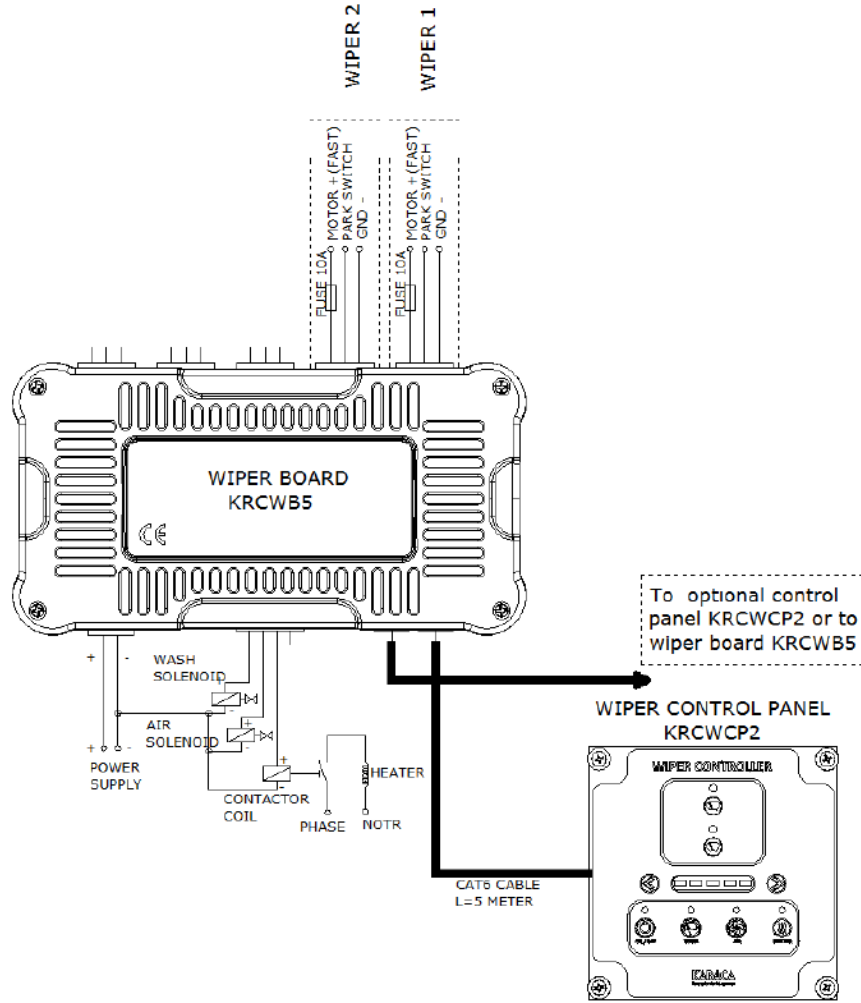
6.7. SYSTEM CONFIGURATION

KRCWB5 board can handle a maximum of 5 wipers. When the number of wipers is above 5, the number of KRCWB5 boards will also increase. The following configurations are available.

No. Of Installed Wipers	Wiper Control Panel	Wiper Board
2	KRCWCP2	KRCWB5
3	KRCWCP3	KRCWB5
4	KRCWCP4	KRCWB5
5	KRCWCP5	KRCWB5
6	KRCWCP6	KRCWB5X2
7	KRCWCP10	KRCWB5X2
8	KRCWCP10	KRCWB5X2
9	KRCWCP10	KRCWB5X2
10	KRCWCP10	KRCWB5X2

*User can connect 2 or more KRCWCPX and KRCWBX according to wiper groups using RJ45 sockets. The recommended connection method is to connect one control panel to the wiper board and not to the other control panel.

6.8. SYSTEM WIRING



7. MAINTENANCE AND CLEANING

External Surface Cleaning: Wipe the device's exterior with a damp cloth. Do not use abrasive cleaning materials.

Periodic Maintenance: The device's mechanical and electronic components should be checked every 6 months.

Wiper Blade Replacement: The wiper arm and blade can be replaced using the details of the Wiper Blade Assembly and Wiper Arm Assembly shown in Figure 13.

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